# Introduction to Network Security

# Network Security

- <u>Subject goals:</u>
- To understand principles of network security:
  - cryptography and its *many* uses beyond "confidentiality"
  - authentication
  - message integrity
  - key distribution and certification
- Security in practice:
  - security in application, transport, network, link layers

- **Textbook:**
  1. Network Security Essentials: Applications and Standards, 3rd Ed. William Stallings
  2. Cryptography and Network Security: Principles and Practices, 4th Ed. William Stallings

# Contents:

1. Cryptography
   - Algorithms and protocols
     - Conventional and public key-based encryption, hash function, digital signatures, and key exchange and distribution
2. Network security applications
   - Applications and tools
     - Kerberos, X.509v3 certificates, PGP, S/MIME, IP security, SSL/TLS, SET, and SNMPv3

# Introduction

- A network consists of two or more devices that are linked in order to share resources or allow communications.
- Security is the act of **protecting** a **person**, **property** or **organization** from an attack.

# Introduction

- Computer Security
  - Generic name for the collection of tools designed to protect data and to thwart hackers
- Network Security
  - Measures to protect data during their transmission

# OSI Security Architecture

- ITU-T X.800 "Security Architecture for OSI"
  - A systematic way of defining and providing security requirements

ITU-T: International Telecommunication Union Telecommunication Standardization Sector

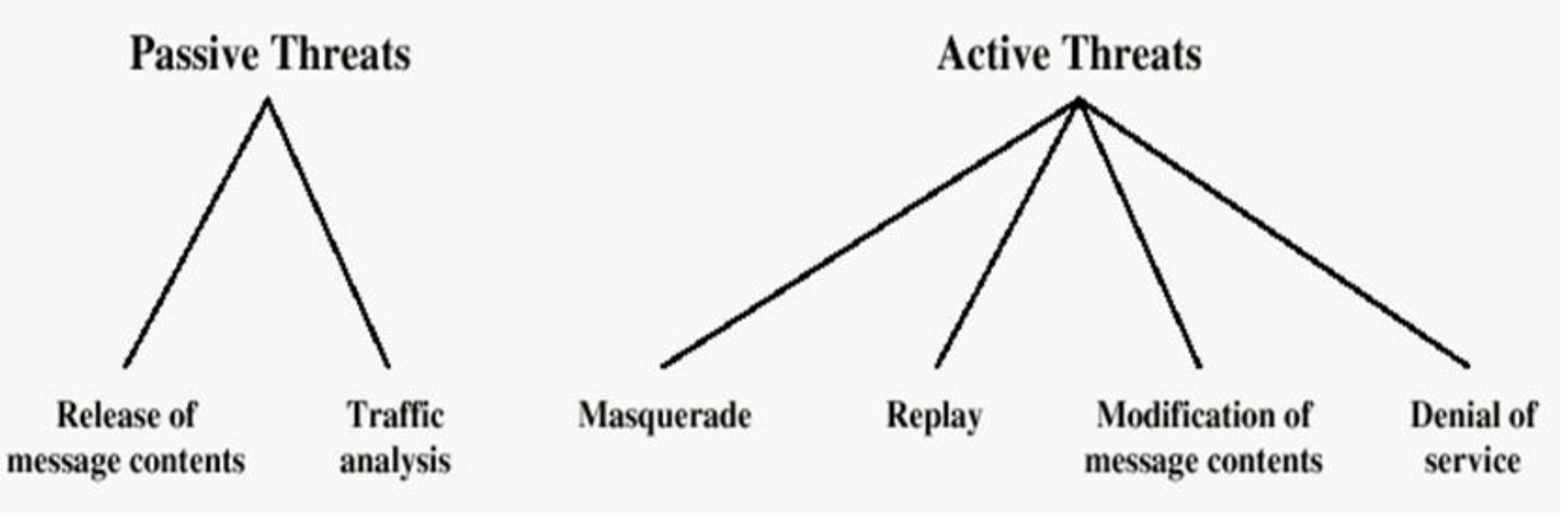
OSI: Open Systems Interconnection

# Aspects of Information Security

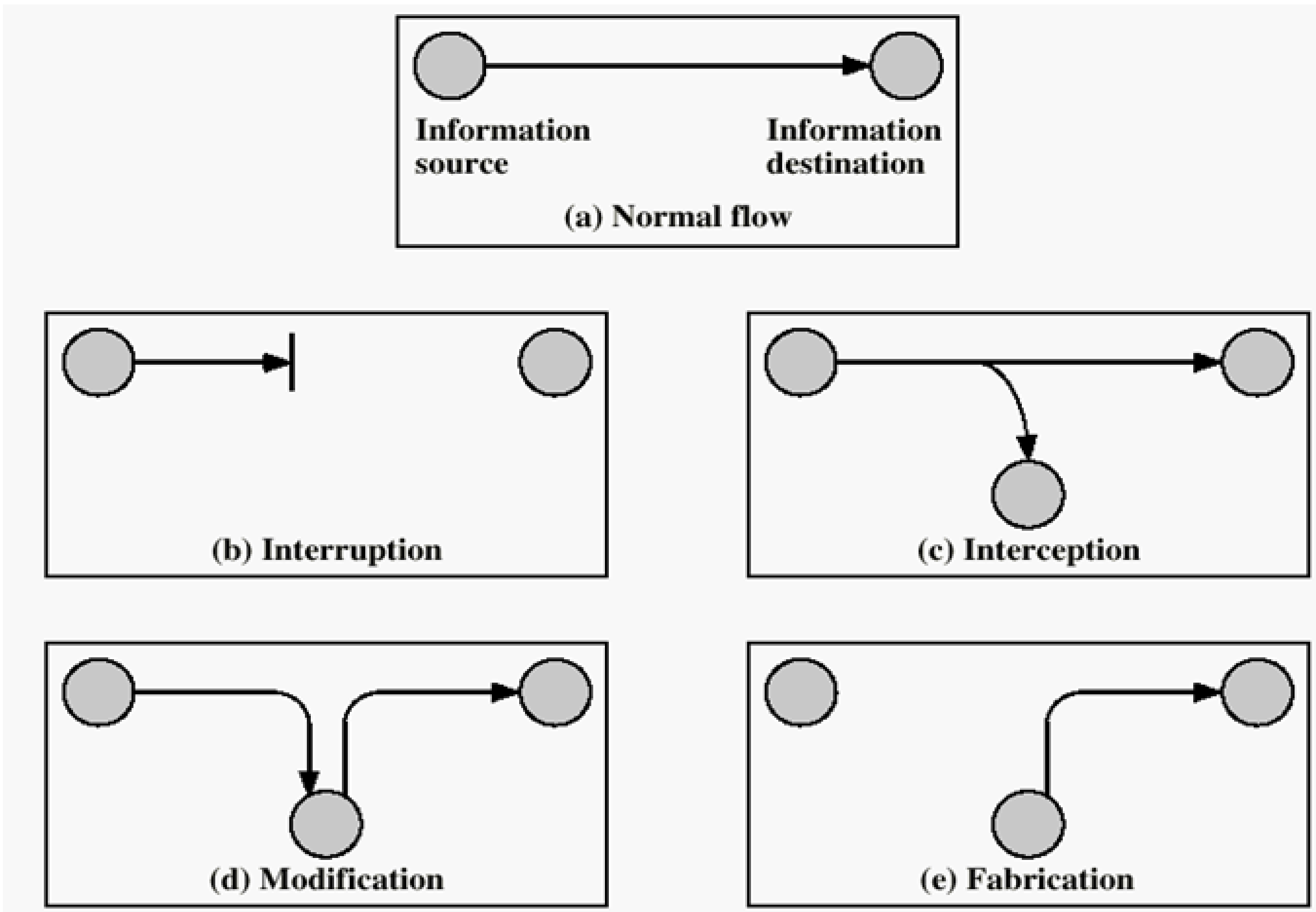
- Security Attack
  - Any action that compromises the security of information.
- Security Mechanism
  - A mechanism that is designed to detect, prevent, or recover from a security attack.
- Security Service
  - A service that enhances the security of data processing systems and information transfers.
    - Makes use of one or more security mechanisms.

# Security Attacks

- There are a wide range of attacks
  - Two generic types of attacks
    - Passive
    - Active



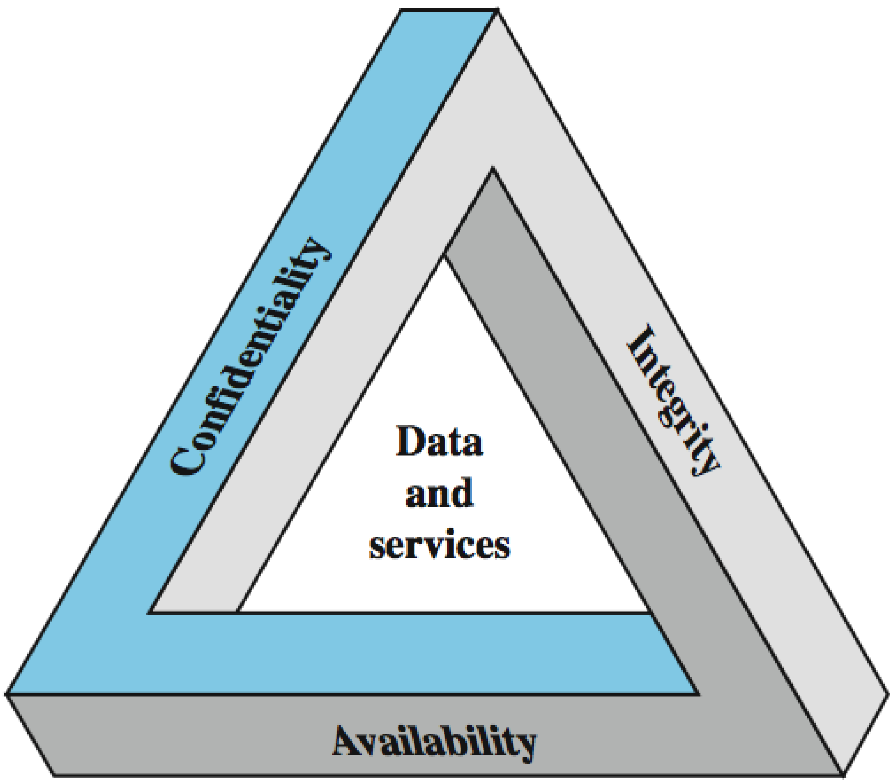
# Security Attack Classification



# Security Attacks

- **Interruption:** This is an attack on availability
- **Interception:** This is an attack on confidentiality
- **Modification:** This is an attack on integrity
- **Fabrication:** This is an attack on authenticity

# Primary Security Goals



# Security Services

X.800

- A service provided by a protocol layer of communicating open systems, which ensures adequate security of the systems or of data transfers

- Confidentiality (privacy)
- Authentication (who created or sent the data)
- Integrity (has not been altered)
- Non-repudiation (the order is final)
- Access control (prevent misuse of resources)
- Availability (permanence, non-erasure)
  - Denial of Service Attacks
  - Virus that deletes files

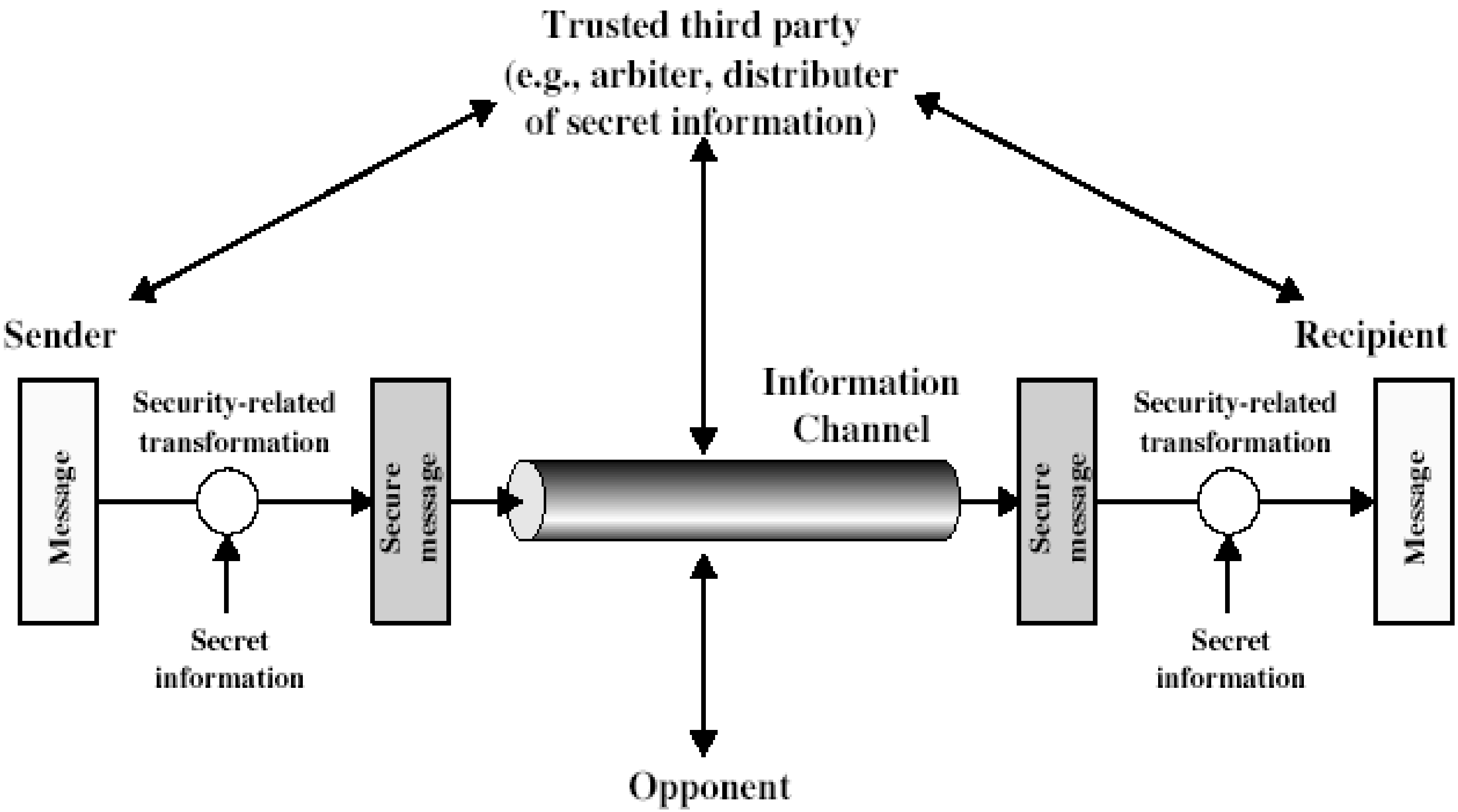
# Security Mechanism

- These are the specific means of implementing one or more security services.
- Features designed to detect, prevent, or recover from a security attack
- No single mechanism that will support all services required
  - **Cryptographic techniques are the basis of almost all security mechanisms**

# Security Mechanisms (X.800)

- Specific security mechanisms:
  - Encipherment, digital signatures, access controls, data integrity, authentication exchange, traffic padding, routing control, notarization
- Pervasive security mechanisms:
  - Trusted functionality, security labels, event detection, security audit trails, security recovery

# Model for Network Security



# Methods of Defense

- Encryption
- Software Controls
  - Limit access in a database or in operating systems
  - Protect each user from other users
- Hardware Controls
  - Smartcard (ICC, used for digital signature and secure identification)
- Policies
  - Frequent changes of passwords
  - Recent study shows controversial arguments
- Physical Controls

# Internet standards and RFCs

- Three organizations in the Internet society
  - Internet Architecture Board (IAB)
    - Defining overall Internet architecture
    - Providing guidance to IETF
  - Internet Engineering Task Force (IETF)
    - Actual development of protocols and standards
  - Internet Engineering Steering Group (IESG)
    - Technical management of IETF activities and Internet standards process

# Conclusion

- Computer networks are typically a shared resource used by many applications representing different interests.
- Unless security measures are taken, a network conversation or a distributed application may be compromised by an adversary.